Physics - Undergraduate Level Assessment 10 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. The best type of laser with which to do spectroscopy over a range of visible wavelengths is
   (a) a dye laser
   (b) a helium-neon laser
   (c) an excimer laser
   (d) a ruby laser

2. Which of the following lasers utilizes transitions that involve the energy levels of free atoms?
   (a) Diode laser
   (b) Dye laser
   (c) Free-electron laser
   (d) Gas laser

3. Electromagnetic radiation emitted from a nucleus is most likely to be in the form of
   (a) gamma rays
   (b) microwaves
   (c) ultraviolet radiation
   (d) visible light

4. The emission spectrum of the doubly ionized lithium atom Li++ (Z = 3, A = 7) is identical to that of a hydrogen atom in which all the wavelengths are
   (a) decreased by a factor of 9
   (b) decreased by a factor of 49
   (c) decreased by a factor of 81
   (d) increased by a factor of 9

5. Which of the following is true about any system that undergoes a reversible thermodynamic process?
   (a) There are no changes in the internal energy of the system.
   (b) The temperature of the system remains constant during the process.
   (c) The entropy of the system and its environment remains unchanged.
   (d) The entropy of the system and its environment must increase.

True / False (10 marks)

Answer True or False

6. True or False: The emission spectrum of the doubly ionized lithium atom Li++ has wavelengths decreased by a factor of 49 compared to hydrogen.
7. True or False: In a reversible thermodynamic process, the entropy of the system and its environment remains unchanged.
8. True or False: The frequency of the next higher harmonic for a closed organ pipe with a fundamental frequency of 131 Hz is 262 Hz.
9. True or False: Cooper pairs in superconductors are primarily attracted to each other through the strong nuclear force.
10. True or False: The eigenvalues of a Hermitian operator can be complex numbers, including imaginary values.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the conditions under which a beam of electrons can diffract when impinging on a crystal surface, and analyze the relationship between the kinetic energy of the electrons and the observed scattering pattern, particularly considering a lattice spacing of 0.4 nm. What is the approximate kinetic energy required to observe this pattern, and why is it significant in the context of electron diffraction?
12. Discuss the implications of replacing the Sun with a black hole of the same mass on the orbits of the planets. What factors contribute to the stability of these orbits?

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